

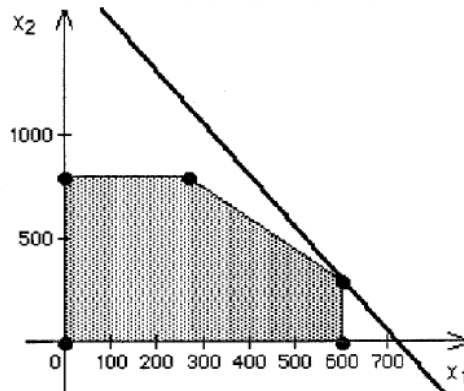
題號 3.1-9

3.1-9.

(a) Let x_1 be the number of units on special risk insurance and x_2 be the number of units on mortgages.

$$\begin{aligned} &\text{maximize } z = 5x_1 + 2x_2 \\ &\text{subject to } \begin{aligned} 3x_1 + 2x_2 &\leq 2400 \\ x_2 &\leq 800 \\ 2x_1 &\leq 1200 \\ x_1 &\geq 0, x_2 \geq 0 \end{aligned} \end{aligned}$$

(b) Optimal Solution: $(x_1^*, x_2^*) = (600, 300)$ and $Z^* = 3600$



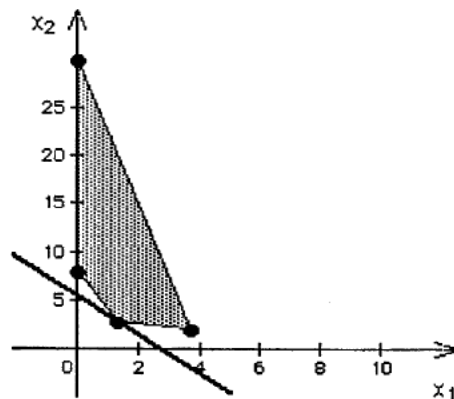
(c) The relevant two equations are $3x_1 + 2x_2 = 2400$ and $2x_1 = 1200$, so $x_1 = 600$ and $x_2 = \frac{1}{2}(2400 - 3x_1) = 300$, $z = 5x_1 + 2x_2 = 3600$.

題號 3.4-8 (a) (b)

3.4-8.

$$\begin{aligned} &\text{(a) minimize } C = 8S + 4P \\ &\text{subject to } \begin{aligned} 5S + 15P &\geq 50 \\ 20S + 5P &\geq 40 \\ 15S + 2P &\leq 60 \\ S, P &\geq 0 \end{aligned} \end{aligned}$$

(b) Optimal Solution: $(S, P) = (x_1^*, x_2^*) = (1.3, 2.9)$ and $C^* = 21.82$



題號 3.4-10(a)

3.4-10.

- (a) Let f_1 = number of full-time consultants working the morning shift (8 a.m.-4 p.m.),
 f_2 = number of full-time consultants working the afternoon shift (Noon-8 p.m.),
 f_3 = number of full-time consultants working the evening shift (4 p.m.-midnight),
 p_1 = number of part-time consultants working the first shift (8 a.m.-noon),
 p_2 = number of part-time consultants working the second shift (Noon-4 p.m.),
 p_3 = number of part-time consultants working the third shift (4 p.m.-8 p.m.),
 p_4 = number of part-time consultants working the fourth shift (8 p.m.-midnight).

$$\text{minimize} \quad C = (40 \times 8)(f_1 + f_2 + f_3) + (30 \times 4)(p_1 + p_2 + p_3 + p_4)$$

$$\begin{aligned} \text{subject to} \quad & f_1 + p_1 \geq 4 \\ & f_1 + f_2 + p_2 \geq 8 \\ & f_2 + f_3 + p_3 \geq 10 \\ & f_3 + p_4 \geq 6 \\ & f_1 \geq 2p_1 \\ & f_1 + f_2 \geq 2p_2 \\ & f_2 + f_3 \geq 2p_3 \\ & f_3 \geq 2p_4 \\ & f_1, f_2, f_3, p_1, p_2, p_3, p_4 \geq 0 \end{aligned}$$

題號 3.4-14(a)

3.4-14.

- (a) Let x_{ij} be the number of tons of cargo type $i = 1, 2, 3, 4$ stowed in compartment $j = F$ (front), C (center), B (back).

$$\begin{aligned} \text{maximize} \quad & P = 320(x_{1F} + x_{1C} + x_{1B}) + 400(x_{2F} + x_{2C} + x_{2B}) \\ & + 360(x_{3F} + x_{3C} + x_{3B}) + 290(x_{4F} + x_{4C} + x_{4B}) \end{aligned}$$

$$\begin{aligned} \text{subject to} \quad & x_{1F} + x_{2F} + x_{3F} + x_{4F} \leq 12 \\ & x_{1C} + x_{2C} + x_{3C} + x_{4C} \leq 18 \\ & x_{1B} + x_{2B} + x_{3B} + x_{4B} \leq 10 \\ & x_{1F} + x_{1C} + x_{1B} \leq 20 \\ & x_{2F} + x_{2C} + x_{2B} \leq 16 \\ & x_{3F} + x_{3C} + x_{3B} \leq 25 \\ & x_{4F} + x_{4C} + x_{4B} \leq 13 \\ & 500x_{1F} + 700x_{2F} + 600x_{3F} + 400x_{4F} \leq 7,000 \\ & 500x_{1C} + 700x_{2C} + 600x_{3C} + 400x_{4C} \leq 9,000 \\ & 500x_{1B} + 700x_{2B} + 600x_{3B} + 400x_{4B} \leq 5,000 \\ & \frac{1}{12}(x_{1F} + x_{2F} + x_{3F} + x_{4F}) - \frac{1}{18}(x_{1C} + x_{2C} + x_{3C} + x_{4C}) = 0 \\ & \frac{1}{12}(x_{1F} + x_{2F} + x_{3F} + x_{4F}) - \frac{1}{10}(x_{1B} + x_{2B} + x_{3B} + x_{4B}) = 0 \\ \text{and} \quad & x_{1F}, x_{2F}, x_{3F}, x_{4F}, x_{1C}, x_{2C}, x_{3C}, x_{4C}, x_{1B}, x_{2B}, x_{3B}, x_{4B} \geq 0 \end{aligned}$$

題號 3.4-15 (a)

3.4-15.

(a) Let x_{ij} be the number of hours operator i is assigned to work on day j for $i = KC, DH, HB, SC, KS, NK$ and $j = M, Tu, W, Th, F$.

$$\begin{aligned} \text{minimize} \quad Z = & 25(x_{KC,M} + x_{KC,W} + x_{KC,F}) + 26(x_{DH,Tu} + x_{DH,Th}) + \\ & 24(x_{HB,M} + x_{HB,Tu} + x_{HB,W} + x_{HB,F}) + \\ & 23(x_{SC,M} + x_{SC,Tu} + x_{SC,W} + x_{SC,F}) + \\ & 28(x_{KS,M} + x_{KS,W} + x_{KS,Th}) + 30(x_{NK,Th} + x_{NK,F}) \end{aligned}$$

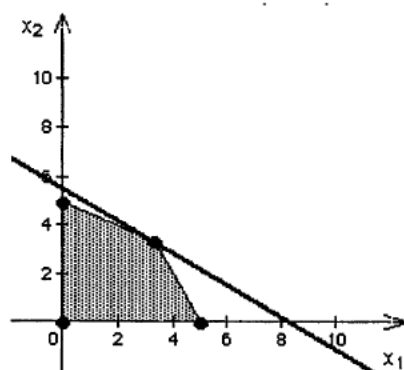
$$\begin{aligned} \text{subject to} \quad & x_{KC,M} \leq 6, x_{KC,W} \leq 6, x_{KC,F} \leq 6 \\ & x_{DH,Tu} \leq 6, x_{DH,Th} \leq 6 \\ & x_{HB,M} \leq 4, x_{HB,Tu} \leq 8, x_{HB,W} \leq 4, x_{HB,F} \leq 4 \\ & x_{SC,M} \leq 5, x_{SC,Tu} \leq 5, x_{SC,W} \leq 5, x_{SC,F} \leq 5 \\ & x_{KS,M} \leq 3, x_{KS,W} \leq 3, x_{KS,Th} \leq 8 \\ & x_{NK,Th} \leq 6, x_{NK,F} \leq 2 \\ & x_{KC,M} + x_{KC,W} + x_{KC,F} \geq 8 \\ & x_{DH,Tu} + x_{DH,Th} \geq 8 \\ & x_{HB,M} + x_{HB,Tu} + x_{HB,W} + x_{HB,F} \geq 8 \\ & x_{SC,M} + x_{SC,Tu} + x_{SC,W} + x_{SC,F} \geq 8 \\ & x_{KS,M} + x_{KS,W} + x_{KS,Th} \geq 7 \\ & x_{NK,Th} + x_{NK,F} \geq 7 \\ & x_{KC,M} + x_{HB,M} + x_{SC,M} + x_{KS,M} = 14 \\ & x_{DH,Tu} + x_{HB,Tu} + x_{SC,Tu} = 14 \\ & x_{KC,W} + x_{HB,W} + x_{SC,W} + x_{KS,W} = 14 \\ & x_{DH,Th} + x_{HB,Th} + x_{NK,Th} = 14 \\ & x_{KC,F} + x_{HB,F} + x_{SC,F} + x_{NK,F} = 14 \\ & x_{ij} \geq 0 \text{ for all } i, j. \end{aligned}$$

題號 3.5-2 (a)(b)

3.5-2.

- (a) maximize $P = 20x_1 + 30x_2$
 subject to $2x_1 + x_2 \leq 10$
 $3x_1 + 3x_2 \leq 20$
 $2x_1 + 4x_2 \leq 20$
 $x_1, x_2 \geq 0$

(b) Optimal Solution: $(x_1^*, x_2^*) = \left(3\frac{1}{3}, 3\frac{1}{3}\right)$ and $P^* = 166.67$



題號 3.5-4 (a) (b)

3.5-4.

- (a) minimize $C = 60x_1 + 50x_2$
 subject to $5x_1 + 3x_2 \geq 60$
 $2x_1 + 2x_2 \geq 30$
 $7x_1 + 9x_2 \geq 126$
 and $x_1, x_2 \geq 0$

(b) Optimal Solution: $(x_1^*, x_2^*) = (6.75, 8.75)$ and $C^* = 842.50$

